

PRECIPITATION EFFECT ON WET-BULBING OUT PROCESS

METEOROLOGIST JEFF HABY

In previous Haby Hints I covered the 1/3 rule and 1/3 - 5 rule for evaporational cooling:

<http://theweatherprediction.com/habyhints/170/>

<http://theweatherprediction.com/winterwx/nowcasting/>

An important consideration to observe is whether the precipitation will be warm rain drops, supercooled raindrops or frozen raindrops. Each of these is defined below and the effect they have on the wet-bulbing out process when temperatures are fairly cool.

WARM RAIN DROPS- These are rain drops whose temperature is above the current surface temperature. The warmth from the rain drops will act to add sensible heat to the surface. In the case of wet-bulb cooling with warm rain drops, use the 1/3 - 5 rule as a guide. Specifically, determine the [dewpoint depression](#), subtract 5 from this number and then divided this new number by 3 then subtract this number from the current temperature. This will give the wet-bulb temperature after evaporational cooling. Example: Suppose the temperature is 35 F with a dewpoint of 23 F. $35 - 23 = 12$, $12 - 5 = 7$, $7/3 = \text{about } 2$. Thus the wet-bulb is 33 F. Warm raindrops can warm the temperature or prevent it from falling further even after the air has saturated.

SUPERCOOLED PRECIPITATION- Supercooled raindrops are liquid raindrops that have a temperature below freezing. Without a surface to freeze on, liquid can cool below the freezing point. Supercooled raindrops hitting the surface will produce freezing rain if the surface is at freezing or below. Even though the freezing process releases [latent heat](#), often the raindrops will be colder than the surface temperature to start off with. In a case with supercooled raindrops, use the 1/3 rule as a guide.

FROZEN PRECIPITATION- Frozen precipitation can take the form of snow, sleet or some hybrid of the two. If temperatures are above freezing, the melting process will help cool the surface. Frozen precipitation can cool the surface temperature in the absence of evaporative cooling. The ice can bring colder air from aloft. In a case with frozen precipitation, use the 1/3 rule as a guide.

Some other wet-bulb cooling considerations:

1. The wet-bulbing out process is most accurate when precipitation is moderate or heavy. Often light precipitation will leave a dewpoint depression of 5 or greater even after hours of precipitation. Thus, the maximum amount of wet-bulb cooling does not occur with light precipitation.
2. Weather sensors have more error with calculating dewpoint than temperature, especially at

cold temperatures. If the dewpoint reading is off a few degrees, the wet-bulb temperature will be a little off also. If the dewpoint depression is less than 4 F, you can often assume the air is already saturated and no more wet-bulb cooling will occur.

3. The 1/3 rule increases in error at very cold temperatures and warm temperatures. Keep in mind the relationship between the amount of moisture in saturated air and the temperature is an exponential relationship. Thus, the cooling will be more than 1/3 at warm temperatures (especially above 80 F) and less than 1/3 at very cold temperatures (especially below 30 F).